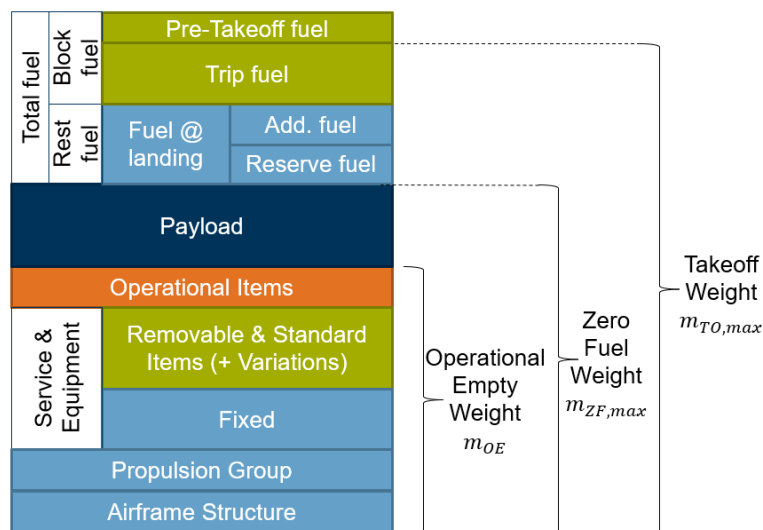


## Solution Tutorial V – Payload-Range Diagram

📖 What is the difference between  $m_{OE}$ ,  $m_{TO,max}$  and  $m_{ZF,max}$ ?

- $m_{OE}$ : the weight of the aircraft as built by the manufacturer + operator items
- $m_{ZF,max}$ : maximum weight of the aircraft before fuel is added ( $=m_{OE} + m_{PL,max}$ )
- $m_{TO,max}$ : maximum weight of the aircraft for take-off ( $=m_{ZF,max} + m_{fuel@maxPL}$ )



Weight definition; based on (Torenbeek, E. (1982), *Synthesis of Subsonic Aircraft Design*, Ch. 8.2), **not scaled!**

📖 Derive the Breguet equations for range and endurance for a turbojet in flight condition I (constant airspeed and lift coefficient).

$$dW = -\dot{m}_{fuel} \cdot g \cdot dt \quad \left| \text{For turbojets: } TSFC = \frac{\dot{m}_{fuel}}{T} \Leftrightarrow \dot{m}_{fuel} = TSFC \cdot T \right.$$

$$\Leftrightarrow dW = -TSFC \cdot T \cdot g \cdot dt \quad \left| \text{Levelled flight: } T = D = \frac{C_D}{C_L} \cdot W \right.$$

$$\Leftrightarrow dW = -TSFC \cdot \frac{C_D}{C_L} \cdot W \cdot g \cdot dt$$

$$\Leftrightarrow dt = -\frac{1}{g \cdot TSFC} \cdot \frac{C_L}{C_D} \cdot \frac{dW}{W} \quad \text{and, knowing that } ds = V \cdot dt: \quad ds = -\frac{V}{g \cdot TSFC} \cdot \frac{C_L}{C_D} \cdot \frac{dW}{W}$$

Integration for flight condition I ( $V = \text{const.}$  &  $C_L = \text{const.}$ ):

$$\Leftrightarrow \Delta t = \int_{W_1}^{W_2} -\frac{1}{g \cdot TSFC} \cdot \frac{C_L}{C_D} \cdot \frac{dW}{W}$$

$$\Leftrightarrow -\frac{1}{g \cdot TSFC} \cdot \frac{C_L}{C_D} \cdot (\ln(W_2) - \ln(W_1)) = -\frac{1}{g \cdot TSFC} \cdot \frac{C_L}{C_D} \cdot \ln \frac{W_2}{W_1}$$

$$\Rightarrow \Delta t = \frac{1}{g \cdot TSFC} \cdot \frac{C_L}{C_D} \cdot \ln \frac{W_1}{W_2} \quad \text{and} \quad \Delta s = \frac{V}{g \cdot TSFC} \cdot \frac{C_L}{C_D} \cdot \ln \frac{W_1}{W_2}$$